C04 -AT2 -CASE STUDY

Q1 Green Data Center Energy Management System

50 marks

A large organization operates a data center containing servers, cooling systems, and renewable-energy equipment and wants to reduce energy consumption and carbon emissions. Current issues include high server energy consumption, inefficient cooling, no distinction between renewable and non-renewable energy, difficulty adding new energy sources, and poor dynamic management. Develop a sustainable C++ OOP system using inheritance, multiple inheritance, virtual base classes, abstract classes, constructors and destructors, composition, pointers, this pointer, and runtime polymorphism. Design abstract DataCenterComponent with Server, CoolingUnit, and RenewableSource. Create GreenServer inheriting from Server and RenewableSource. Use virtual inheritance and an EnergyMeter member class. Implement pure virtual calculateEnergy(). Server uses CPU energy + memory energy; CoolingUnit uses cooling energy; RenewableSource calculates renewable energy generated. Demonstrate DataCenterComponent*, new, delete, overriding, this pointer, and a menu-driven program.

Your Code

```
#include <iostream>
using namespace std;
class DataCenterComponent {
protected:
    string name;
public:
    DataCenterComponent(string n) { this->name = n; }
    virtual double calculateEnergy() = 0;
    virtual void display() = 0;
    virtual ~DataCenterComponent() {}
};
class EnergyMeter {
public:
    double energy;
    EnergyMeter(double e) { energy = e; }
};
class Server : virtual public DataCenterComponent {
protected:
    double cpu, memory;
    EnergyMeter *meter;
public:
    Server(string n, double c, double m)
        : DataCenterComponent(n), cpu(c), memory(m) {
        meter = new EnergyMeter(c + m);
```

Input

4

Output

Visible Test Cases

Test Case 1

Q2 **Carbon-Aware Cloud Computing Management System** 50 marks

A cloud service provider is modernizing its Carbon-Aware Cloud Management System to reduce the environmental impact of computing workloads. The system manages AI Workloads, Web Applications, and Data Analytics Workloads. Current issues include high energy consumption, inflexible energy-management strategies, difficulty supporting new workload categories, inefficient resource management, and inflexible carbon-emission calculation. Develop a C++ OOP application using inheritance, virtual base classes, abstract classes, constructors in derived classes, composition, pointers, the this pointer, pointers to derived classes, and runtime polymorphism. Design an abstract base class Workload with AIWorkload, WebWorkload, and AnalyticsWorkload. Create CriticalWorkload using multiple inheritance from AIWorkload and AnalyticsWorkload, use virtual base classes to resolve ambiguity, and include an EnergyMonitor member class. Implement pure virtual calculateCarbon(). Demonstrate Workload*, new, delete, this pointer, pointer to derived class, and a menu-driven program.

Your Code

```cpp
#include <iostream>
using namespace std;
class Workload {
protected:
    string name;
public:
    Workload(string n) {
        this->name = n;
    }
    virtual double calculateCarbon() = 0;
    virtual void display() = 0;
    virtual ~Workload() {}
};
class EnergyMonitor {
public:
    double energy;
    EnergyMonitor(double e) {
        energy = e;
    }
};
class AIWorkload : virtual public Workload {
protected:
    double energy;
```

Input

```
1.AI  2.Web  3.Analytics  4.Critical
Enter choice: 4
```

Output

Visible Test Cases

Test Case 1